

Signals Final Project

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1 introduction

For this project, we created an acoustic modem receiver. An acoustic modem sends a digital signal encoded in sound waves. Specifically, the digital '1' or '0' is encoded in the phase offset of a sinusoidal signal. To ensure the signal we sent was large enough frequency such that it could be transported through the air, we multiplied by a cosine wave that had a carrying frequency that was sufficiently high. After this, we can multiply by the same cosine wave used to convert our binary encoded message to the sinusoidal acoustic waveform, apply a low pass filter, and minor post-processing to recover a signal close to the original.

2 Block Diagram

The block diagram of our entire system can be visualized below. This diagram indicates a theoretical practical implementation for the acoustic modem system, which is not the case we considered with the MVP. For example, in the MVP - as we are not transmitting a signal over the air - there will be no losses due to the $H(j\omega)$ transfer function.

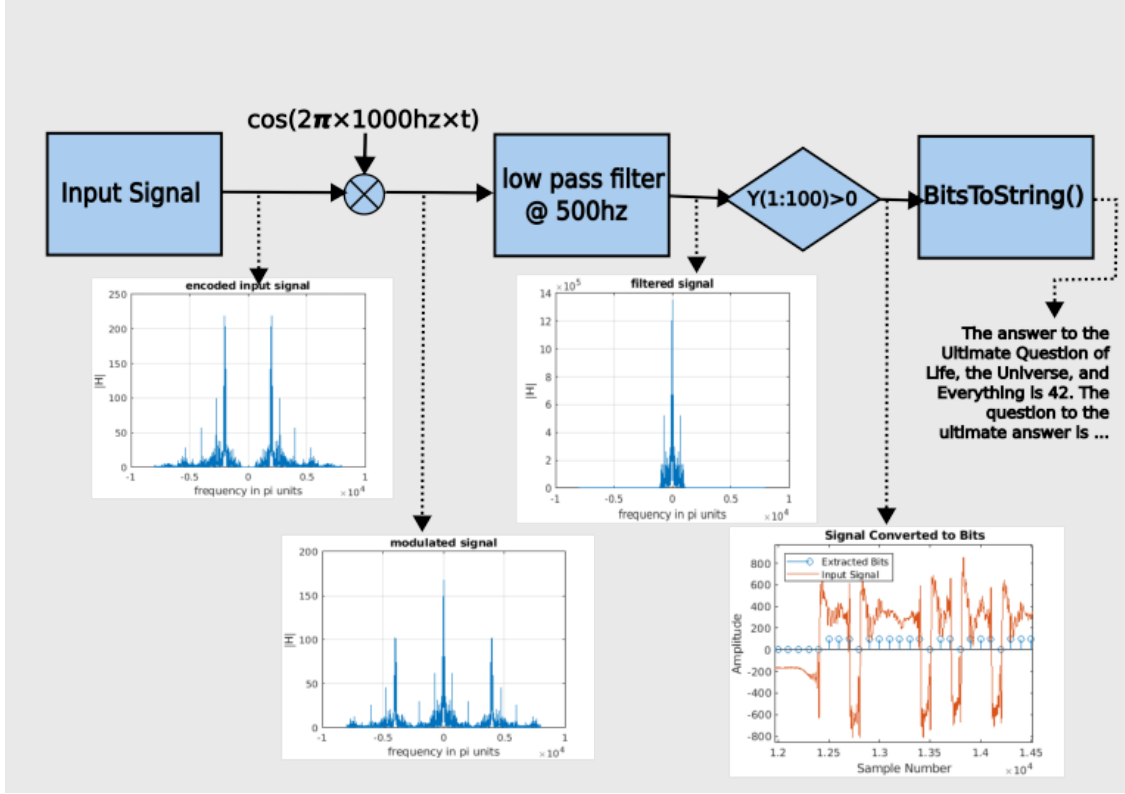


Figure 1: Block diagram of our receiver.

3 Methods

We started with two given signals to decode. One short with the answer of hello and a much longer one, which was more philosophical.

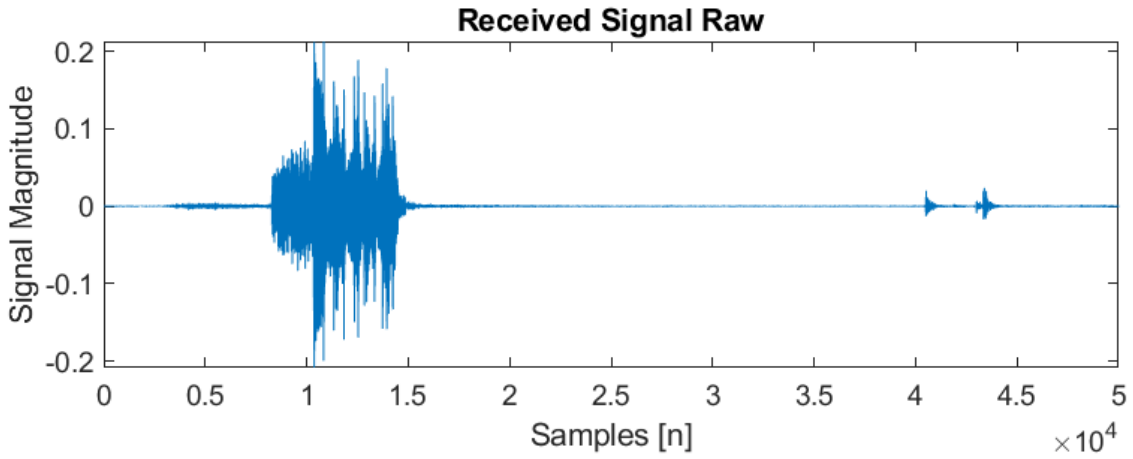


Figure 2: Time domain graph of the raw signal.

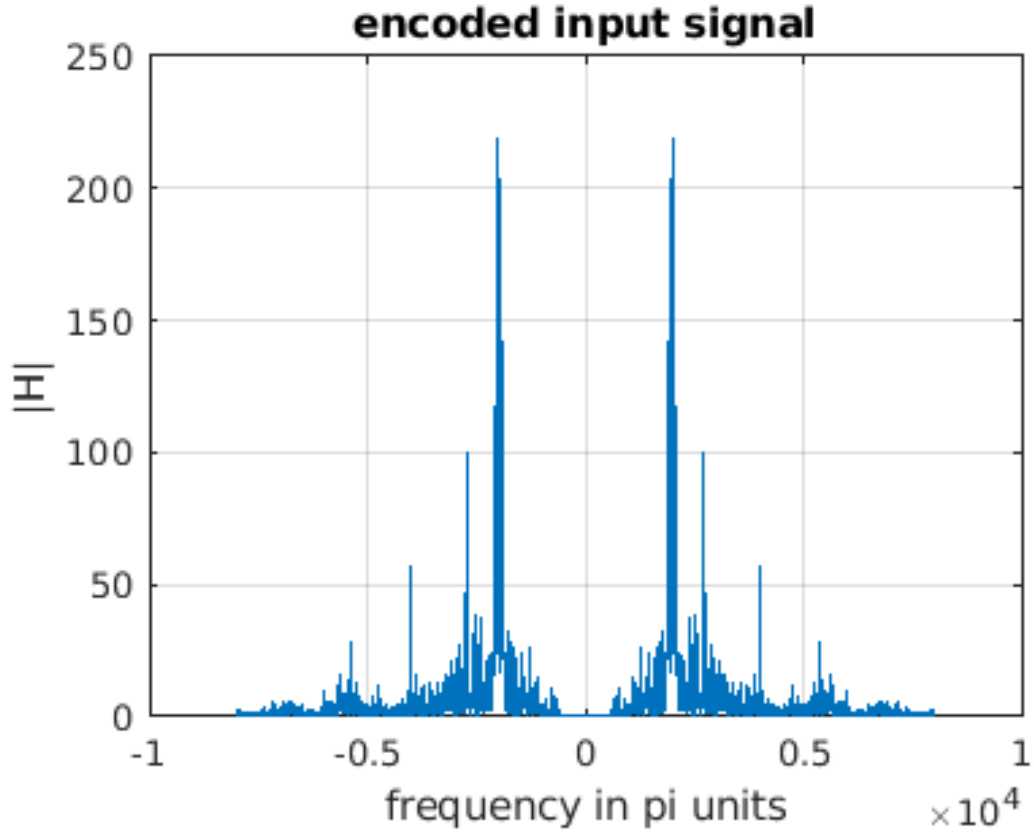


Figure 3: Frequency domain graph of the raw signal.

We first multiplied the signal by another cosine function. This allows us to isolate the signal and later filter it out.

$$\cos(2\pi \cdot 1000 \cdot t); \quad (1)$$

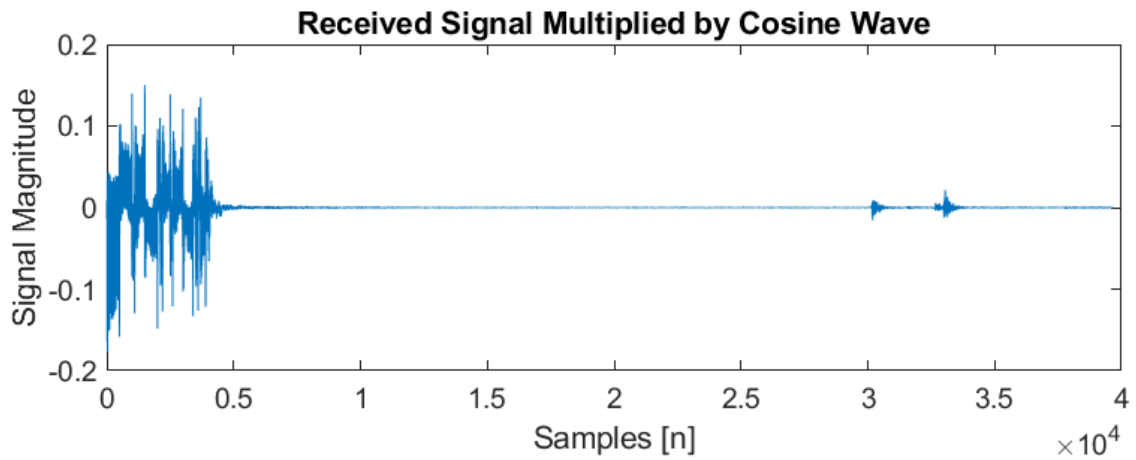


Figure 4: Time domain graph of modulated signal.

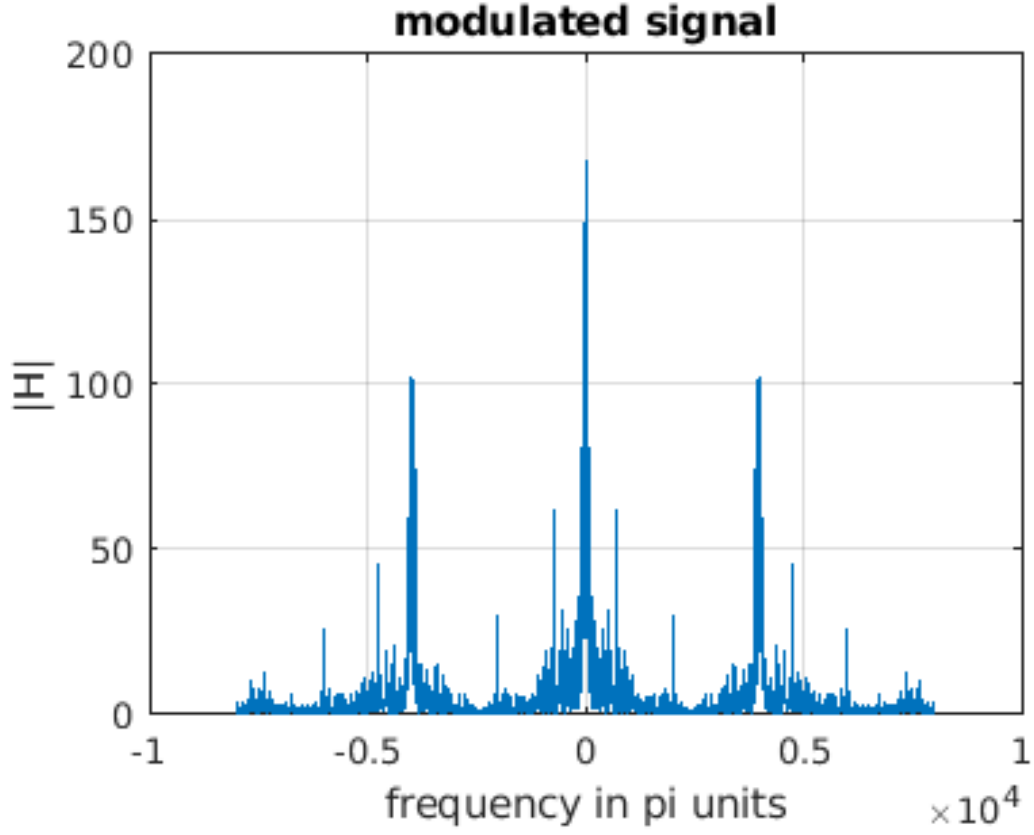


Figure 5: Frequency domain graph of modulated signal.

Once we modulate the signal we then are able to apply a low pass filter to the signal to get rid of any noise and get a clearer reading. Our lowpass sinc filter was at 500Hz.

$$\frac{W}{\pi} \text{sinc}\left(\frac{W}{\pi} \cdot t\right), \quad (2)$$

where $W = 200$.

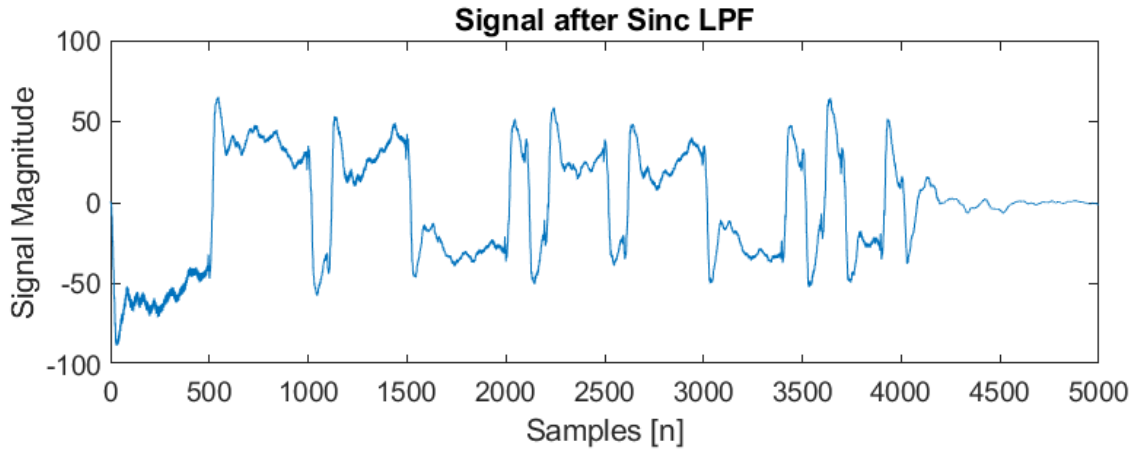


Figure 6: Time domain graph of the filtered signal.

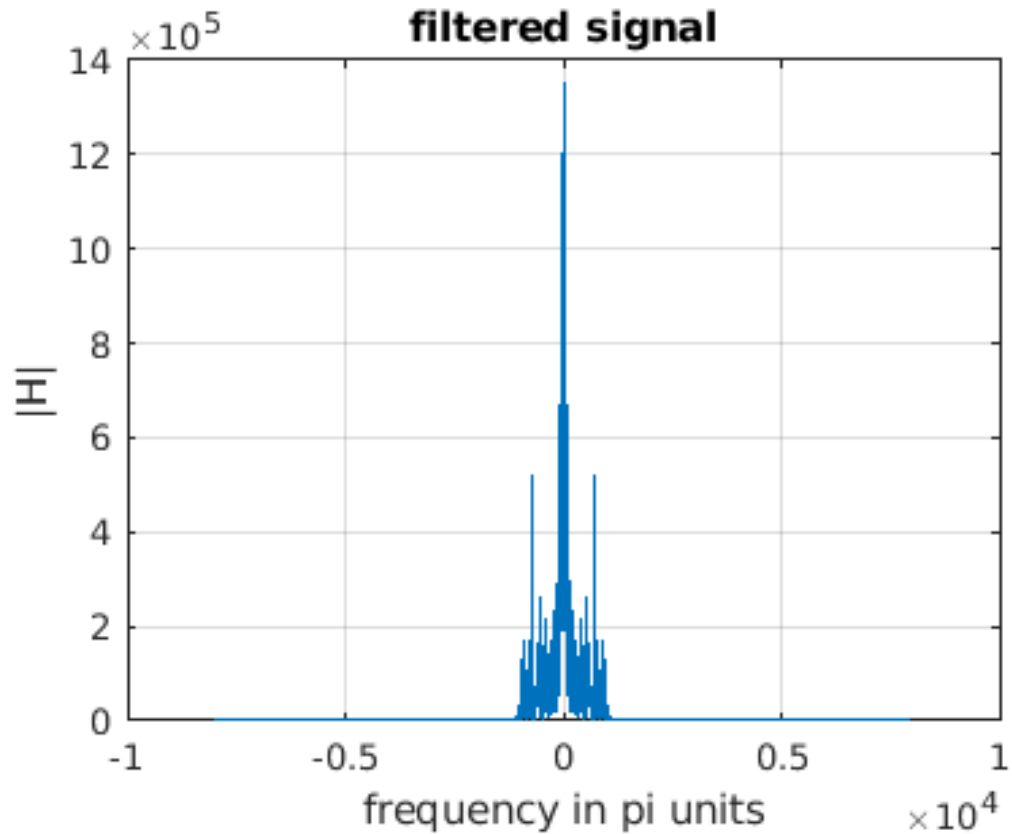


Figure 7: Frequency domain graph of the filtered signal.

We were pretty close to the original signal. You are able to see the rising and falling edges. We only had to convert the signal then to bits to read the message. Below you can see the stem plot of extracted bits and if they are '0' or '1'.

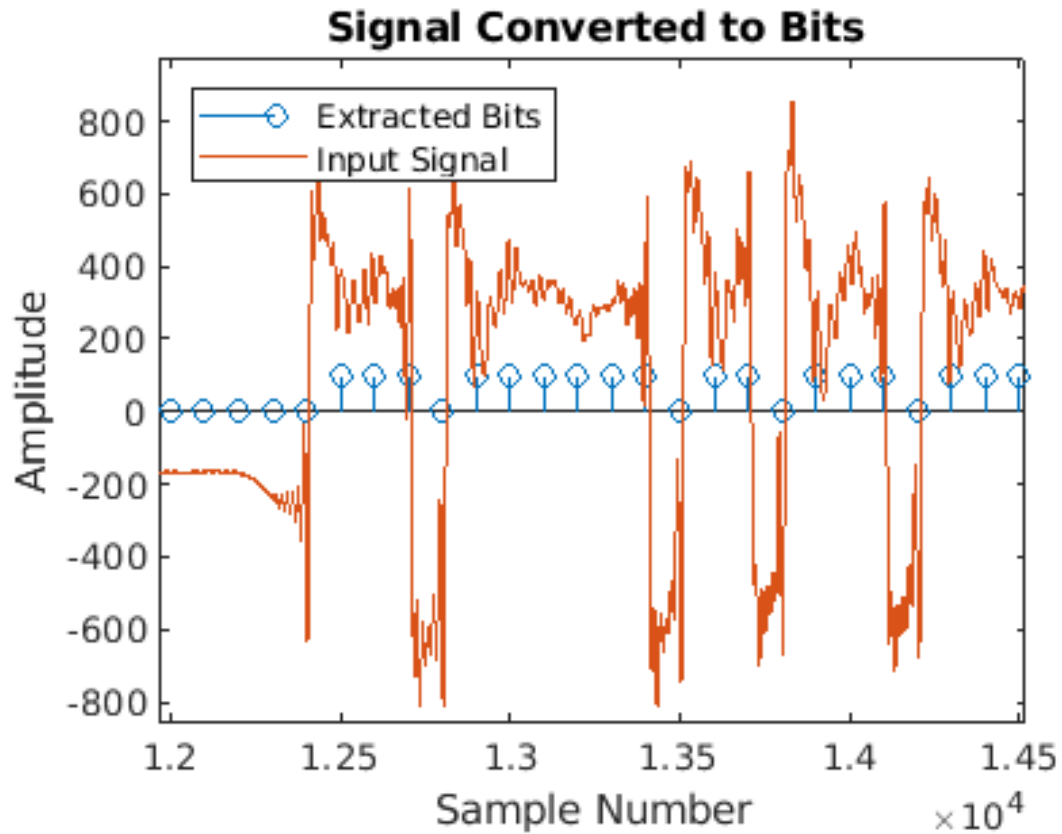


Figure 8: Conversion of the manipulated signal into bits.

4 Results

The short message was: "Hello"

The long message was: "The answer to the Ultimate Question of Life, the Universe, and Everything is 42. The question to the ultimate answer is ..."

5 Video

https://youtu.be/sim5sdm_xu0